

Microsoft

You are a Product Analyst on the Microsoft Teams collaboration platform....

Question 1: For October 2024, what is the average number of chat messages sent per user in Microsoft Teams? This baseline metric will help us understand overall user engagement.

```
SELECT AVG(number_message) AS avg_number_chat_per_user
FROM(
  SELECT user_id, COUNT(message_id) AS number_message
  FROM fct_chat_interactions
  WHERE interaction_date >= '2024-10-01' AND interaction_date
< '2024-11-01'
  AND feature_used IN ('mention','reply')
  GROUP BY user_id) AS total
```



Microsoft

You are a Product Analyst on the Microsoft Teams collaboration platform....

Question 2: In October 2024, which are the top five chat features ranked by the number of unique users interacting with them? Identifying these features will help prioritize enhancements to reduce context-switching.

```
SELECT *
FROM (
    SELECT
        feature_used,
        COUNT(DISTINCT(user_id)) AS num_users,
        DENSE_RANK() OVER(ORDER BY COUNT(DISTINCT(user_id)) DE
SC) AS ranking
    FROM fct_chat_interactions
    WHERE DATE_TRUNC('month', interaction_date)::DATE = '2024-
10-01'
    GROUP BY feature_used
) AS total
WHERE ranking < 6
ORDER BY ranking
LIMIT 5
```



Microsoft

You are a Product Analyst on the Microsoft Teams collaboration platform....

Question 3: For October 2024, what percentage of highly engaged users (those sending more than 50 messages) used the 'reply' feature at least once? This metric directly informs recommendations for feature enhancements that boost engagement and minimize context-switching.

```
WITH top50 AS (  
    SELECT user_id  
    FROM fct_chat_interactions  
    WHERE DATE_TRUNC('month', interaction_date)::DATE = '2024-  
10-01'  
    GROUP BY user_id  
    HAVING COUNT(message_id) >= 50  
)  
reply AS (  
    SELECT COUNT(DISTINCT user_id) AS user_reply  
    FROM fct_chat_interactions  
    WHERE DATE_TRUNC('month', interaction_date)::DATE = '2024-  
10-01'  
    AND feature_used = 'reply'  
    AND user_id IN (SELECT user_id FROM top50)  
)  
top50_count AS (  
    SELECT COUNT(*) AS num_top50  
    FROM top50  
)  
SELECT  
    reply.user_reply,  
    top50_count.num_top50,  
    (reply.user_reply::FLOAT / top50_count.num_top50) * 100 AS  
percentage  
FROM reply, top50_count;
```

